

2201 321 Veterinary Virology

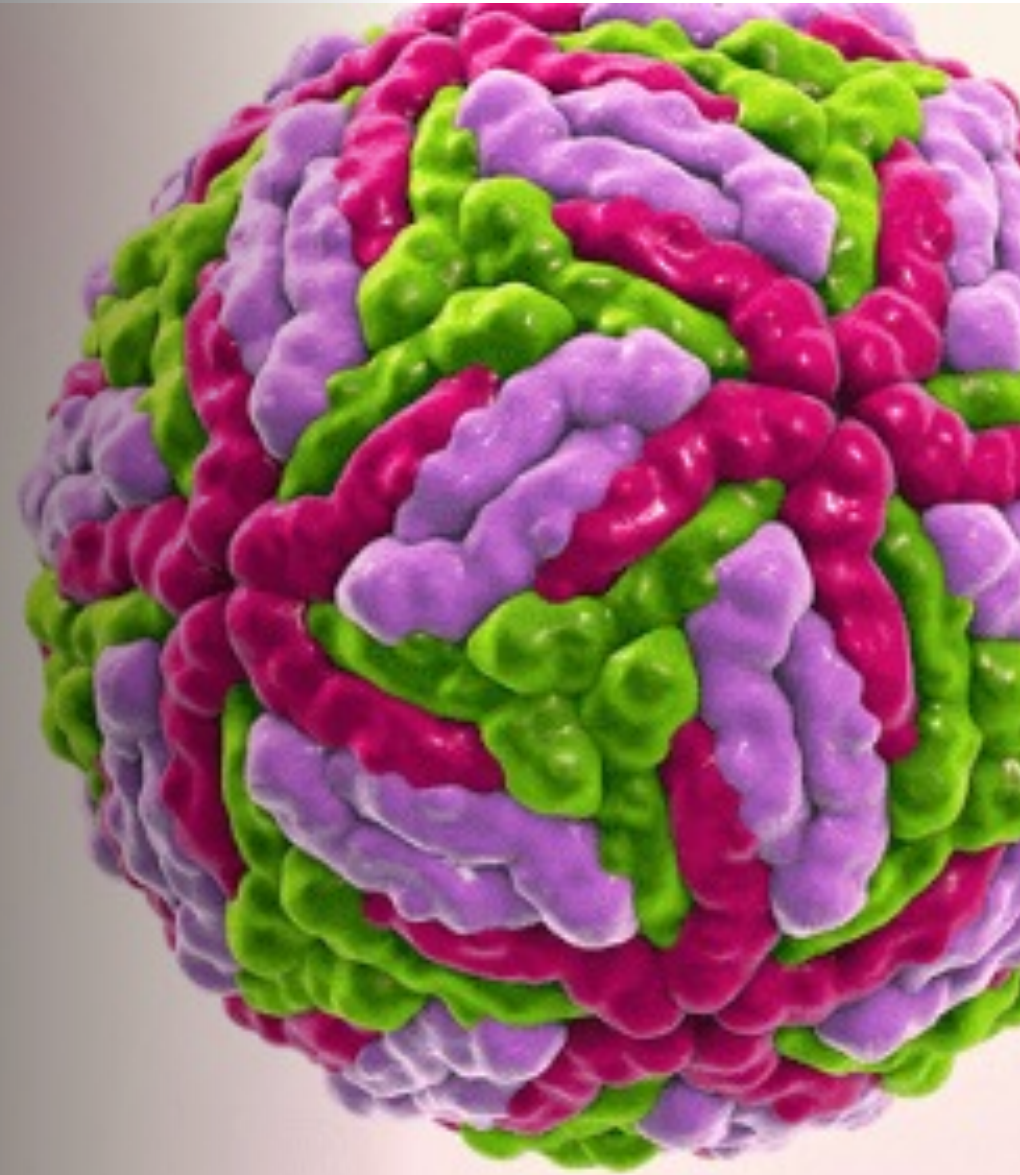


Family *Togaviridae* and Family *Flaviviridae*

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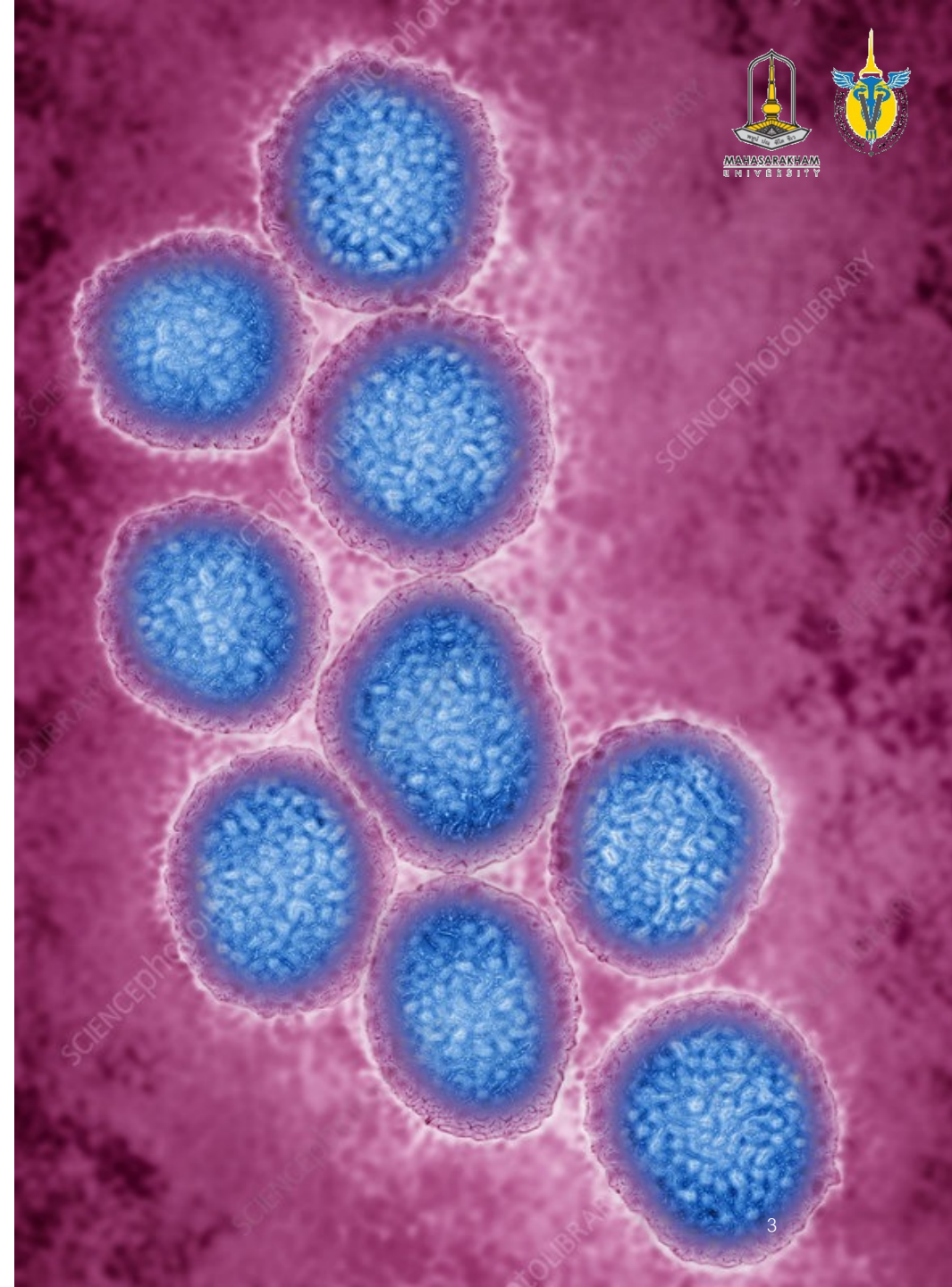


วัตถุประสงค์

- นิสิตมีความรู้ความเข้าใจเกี่ยวกับไวรัสและโรคที่เกิดจากไวรัสใน Family *Togaviridae* และ Family *Flaviviridae* ดังนี้
 - ลักษณะและชีววิทยาของไวรัส
 - การก่อโรคในสัตว์ พยาธิกำเนิดและอาการแสดงทางคลินิก
 - ตัวอย่างเพื่อชั้นสูตรทางห้องปฏิบัติการ วิธีการ และเทคนิคที่ใช้ชั้นสูตรทางห้องปฏิบัติการ
 - การควบคุมและการป้องกันโรค

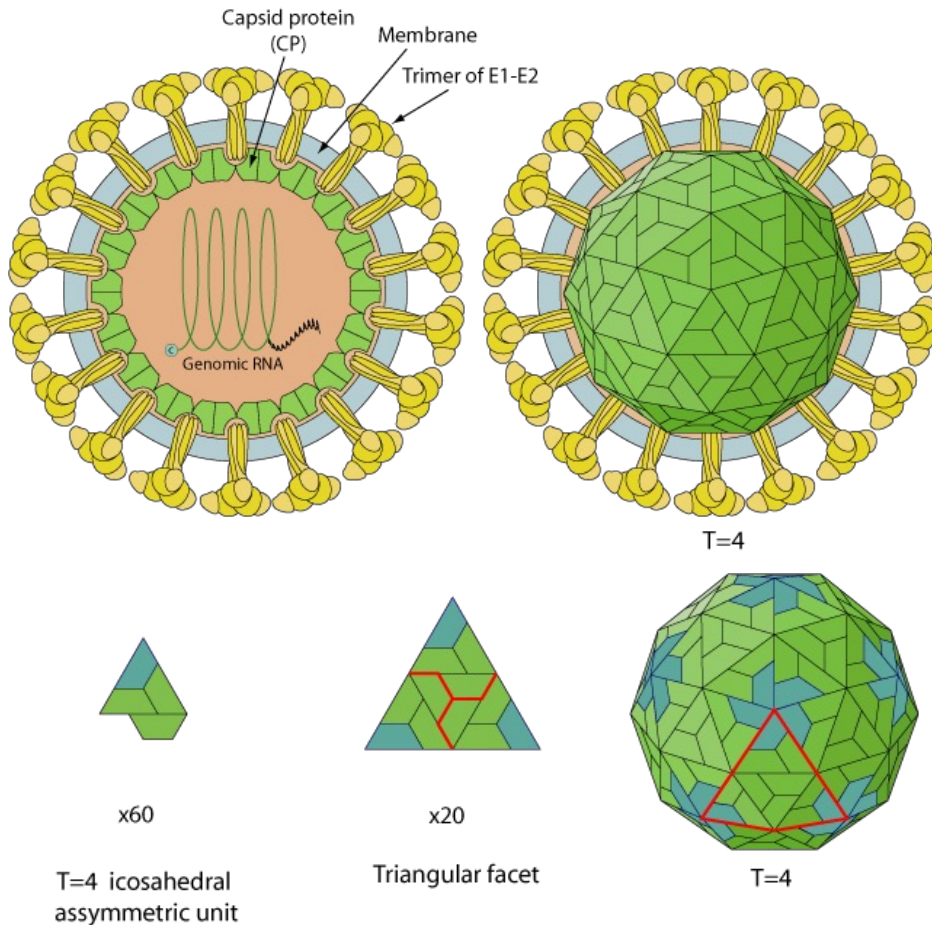
Family *Togaviridae*

- The genera in family *Togaviridae* -
 - Genus *Alphavirus*; includes many species that are mostly mosquito-borne and pathogenic in their vertebrate hosts
 - Chikungunya virus
 - Equine alphaviruses
 - Eastern equine encephalitis virus
 - Highlands J virus
 - Venezuelan equine encephalitis virus
 - Western equine encephalitis virus
 - Salmon pancreas disease virus – infect only fish
 - Before April 2019, the family also contained the genus *Rubivirus* that has now been moved to the family *Matonaviridae*



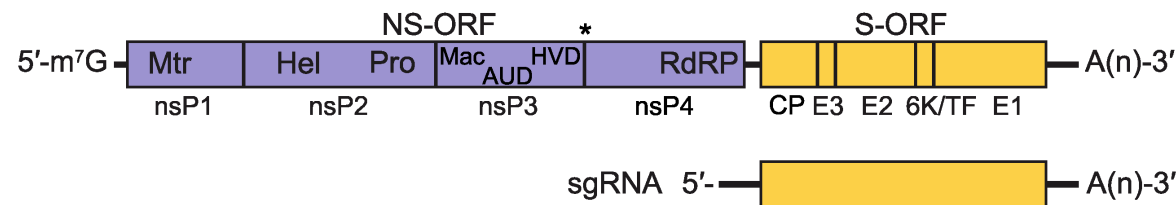
Virion properties

- Virions are spherical, enveloped particles about 70 nm in diameter
- The particle consists of a nucleocapsid core surrounded by a lipid bilayer that is embedded with glycoprotein spikes
 - The nucleocapsid core comprises 240 copies of capsid protein that surround the viral genome
 - The lipid bilayer is host-derived from the site of budding
 - Alphaviruses bud from the plasma membrane although, in arthropods, budding into internal vesicles has also been observed
 - Eighty trimeric glycoprotein spikes cover the surface of alpha-virions, each spike is composed of three E1-E2 heterodimers
- The genome is a single-strand of positive-sense RNA that is capped and has a poly-A tail
- Virions can be physically denatured by treating with various chemicals including urea, formaldehyde, beta-propiolactone, detergents, and acids
 - Infectivity is also decreased by heat inactivation, by exposure to low pH in the absence of lipid vesicles and exposure to ultraviolet light



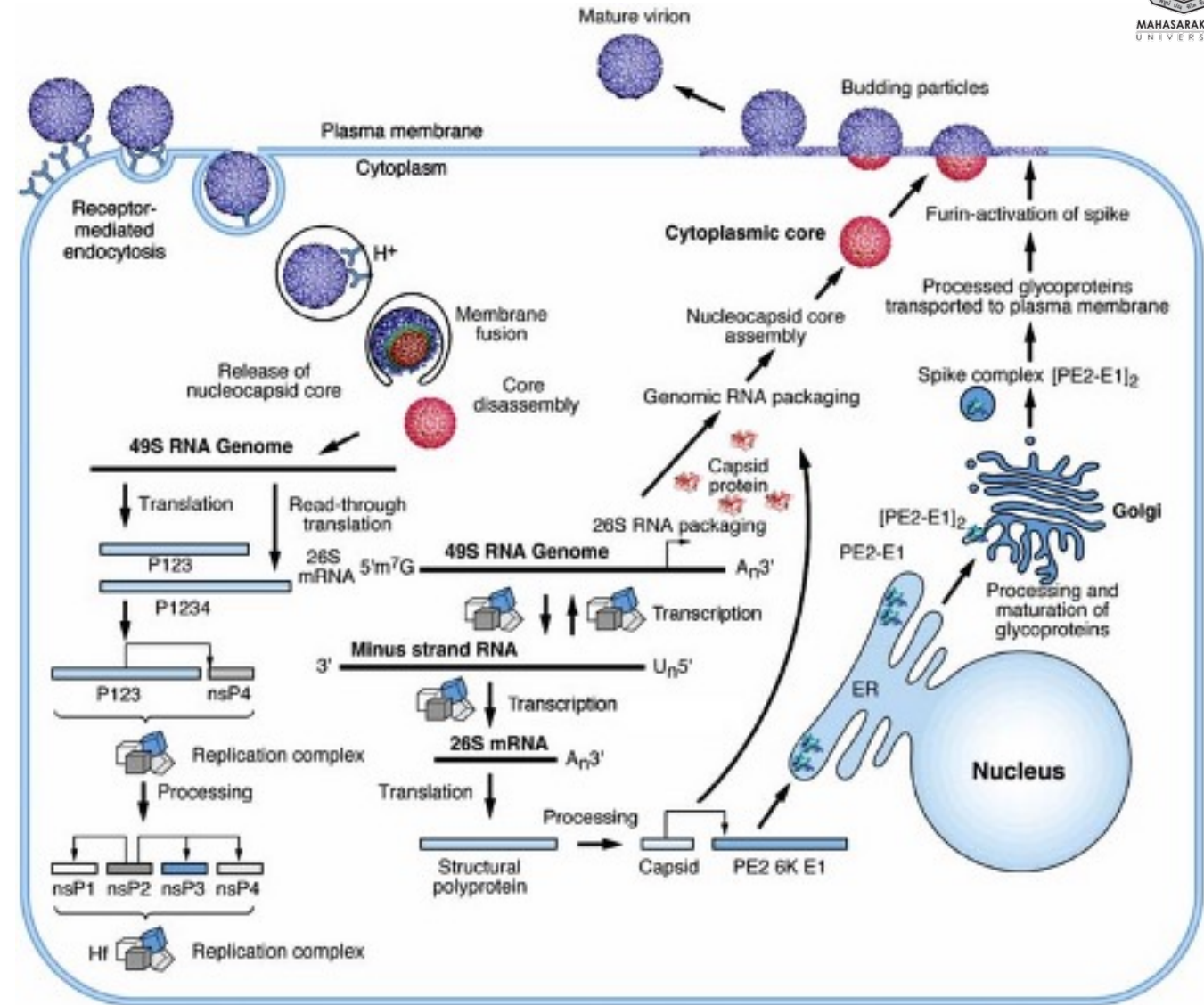
The genome and viral proteins

- The genome of alphaviruses encodes 10 different proteins
 - Non-structural proteins (nsP1-nsP4); important for replicating the viral genome
 - Structural proteins (capsid, E3, E2, 6K/TF, and E1); function in virus assembly
 - The structural proteins are translated as a polyproteins from the subgenomic mRNA - host protease signalase cleaves the individual proteins from the polyproteins
 - The spikes formed by E2 and E1 are responsible for viral entry
 - E2 protein binds to the host-cell receptor and interacts with the capsid protein
 - E1 protein is a class II fusion protein that mediates fusion between the virus membrane and the host cell membrane in the endosome
 - The E3 stabilizes the E2/E1 heterodimers and trimers during spike assembly
 - The exact roles of 6K and TF are not known but 6K has been shown to have viroporin activity and TF has been shown to be a virulence factor

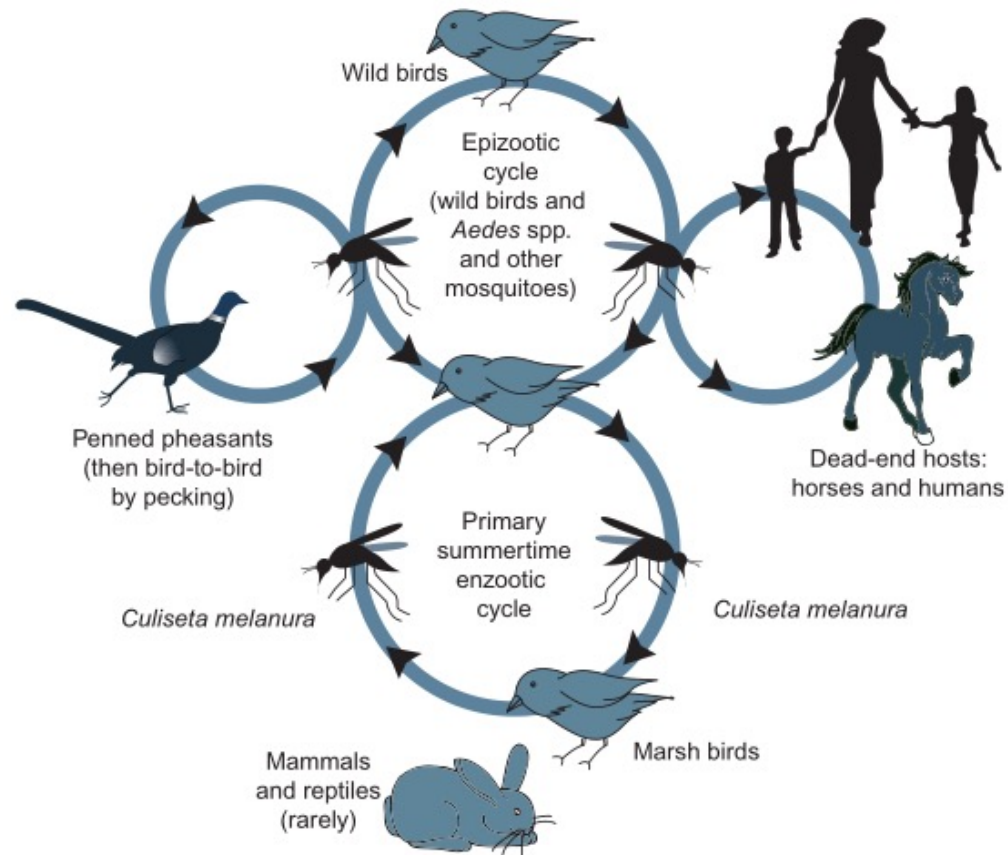


Virus replication

- Replication; cytoplasmic, in vesicles derived from the plasma membrane/endosomal compartment
- Translation; non-structural proteins are translated from genomic RNA, and structural proteins from subgenomic RNA, both as polyprotein precursors
- Assembled virions bud from plasma membrane



Genus *Alphavirus*



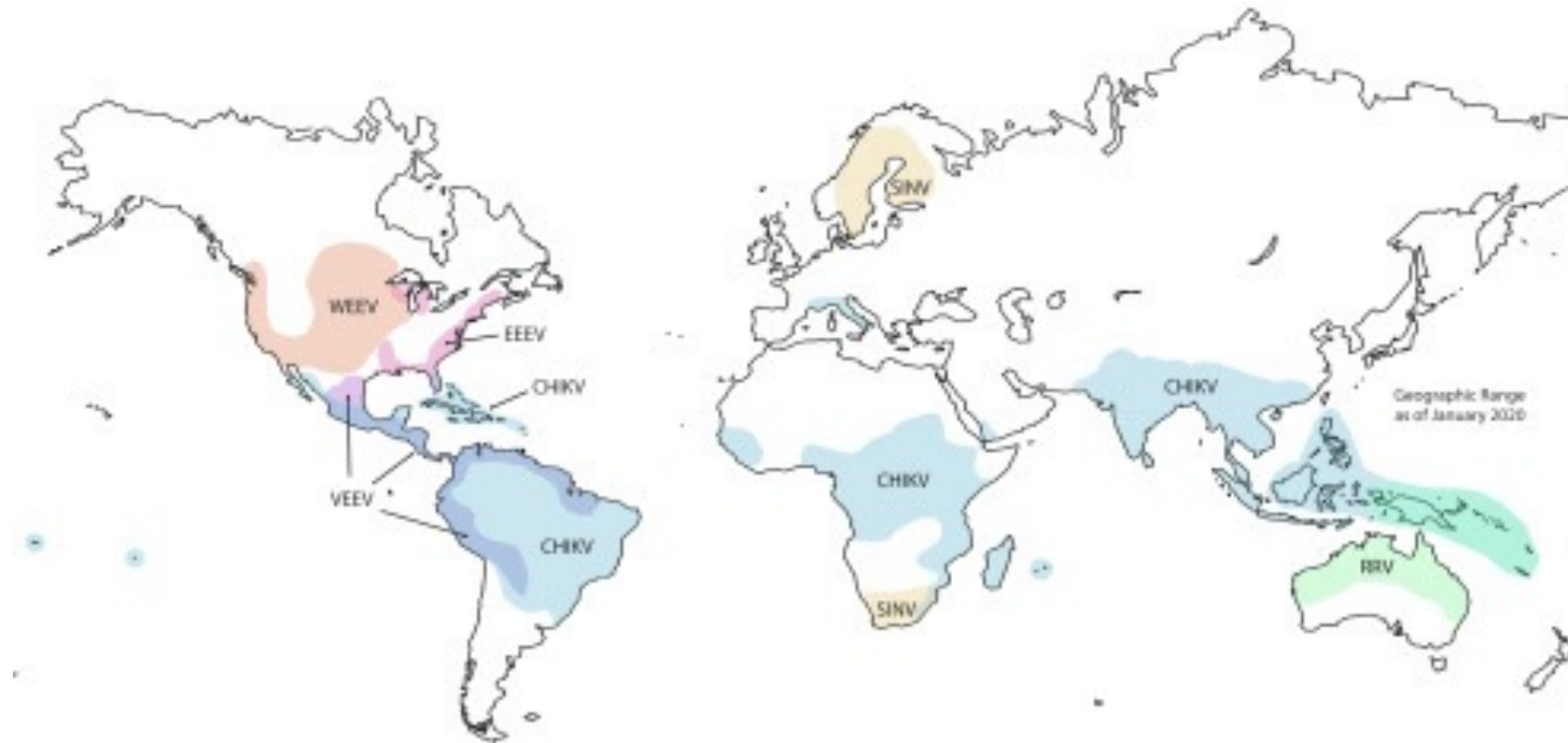
- **Arthropod-borne viruses**

- Equine alphaviruses

- Eastern equine encephalitis virus (EEEV)
 - Highlands J virus
 - Venezuelan equine encephalitis virus (VEEV)
 - Western equine encephalitis virus (WEEV)

- Chikungunya virus

Global geographical distribution of *Alphaviruses*



- Alphaviruses have a global geographical distribution, present on every continent except Antarctica, though their specific locations vary by species. Several viruses are found in tropical and subtropical regions, such as Chikungunya (Asia, Africa, the Americas), and specific viruses have unique distributions, with Eastern Equine Encephalitis (EEE) and Western Equine Encephalitis (WEE) concentrated in North and South America.

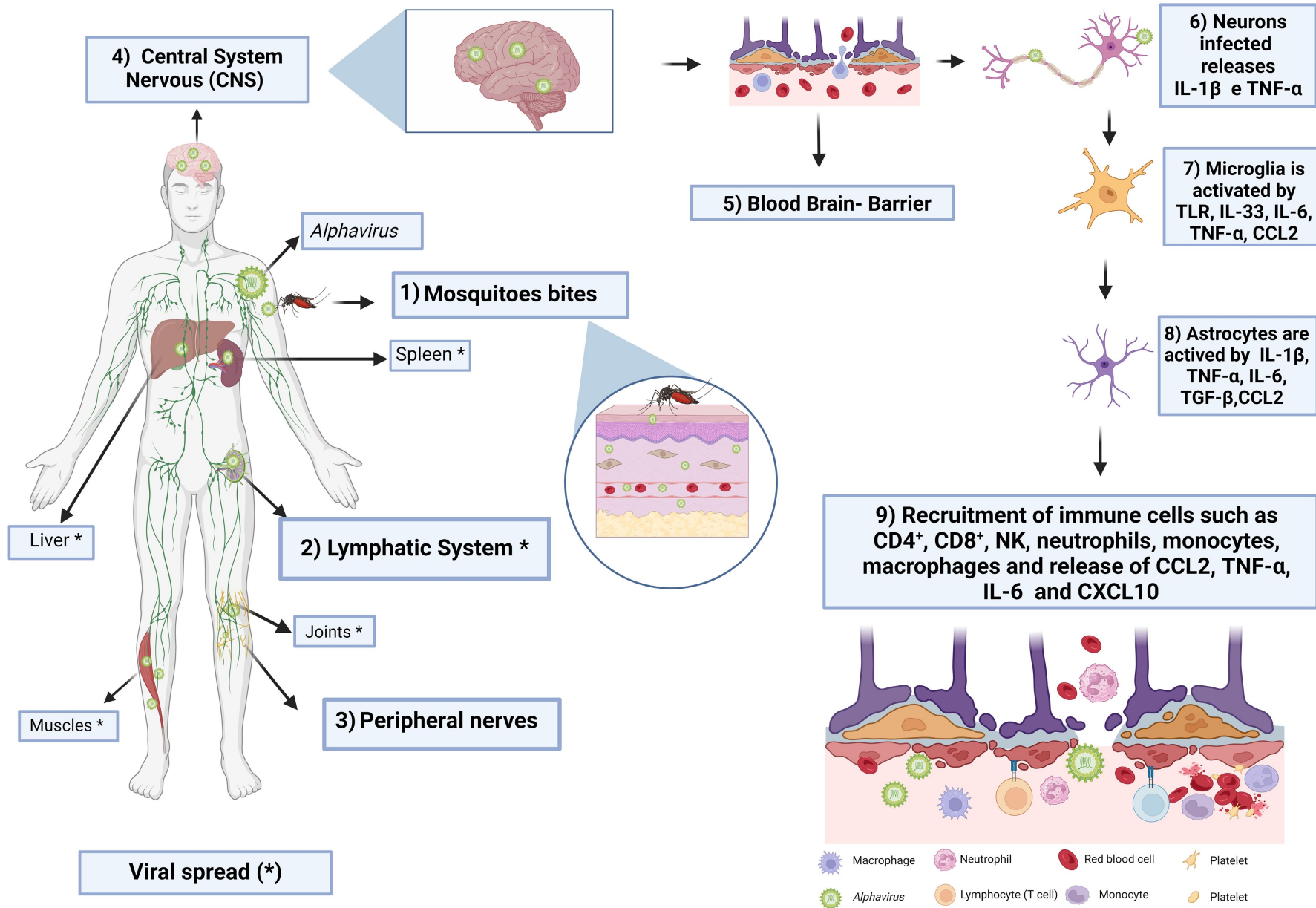
Equine alphaviruses

- Equine alphavirus species
 - Eastern equine encephalitis virus (EEEV)
 - Highlands J virus
 - Venezuelan equine encephalitis virus (VEEV)
 - Western equine encephalitis virus (WEEV)
- Severe disease in horses and humans*
- Transmission cycles
 - Insect vector: mosquitoes
 - Primary host affected –
 - Wild birds (asymptomatic carriers)
 - Horses, pheasants, and humans (severe, often fatal symptoms)



Pathogenesis

- The virus enters via mosquito bite and replicates in lymphoid tissues
- Spreads to the bloodstream; virus damages blood blood-brain barrier
- Reaches the CNS – virus replication in the brain causing encephalitis
 - Inflammatory response exacerbates neural tissue damage, and death
 - The involvement of microglia and astrocytes leads to further damage
- Host immune response: stimulation of macrophages, dendritic cells, T lymphocytes, and B lymphocytes
 - The immune response facilitates inflammation, which leads to tissue and neural damage
- Outcome for survivors – neurological complications or death in severe cases
- Viral spread to human; horses can contribute to virus spread when viremia levels are high enough for mosquitoes to transmit the virus



Clinical Features

- **Subclinical** – transient fever, anorexia, depression, tachycardia
- **Progressive systemic disease** – virus access to CNS after incubation period with 4-6 days, high fever, incoordination and neurologic manifestations
- **Neurologic manifestations** – such as drooping ears, flaccid lips, irregular gait, circling, and encephalitis
- **Terminal stages** – inability to rise, paralysis, convulsions



Diagnosis

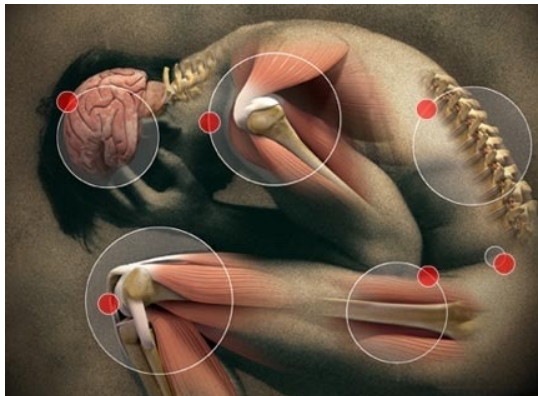
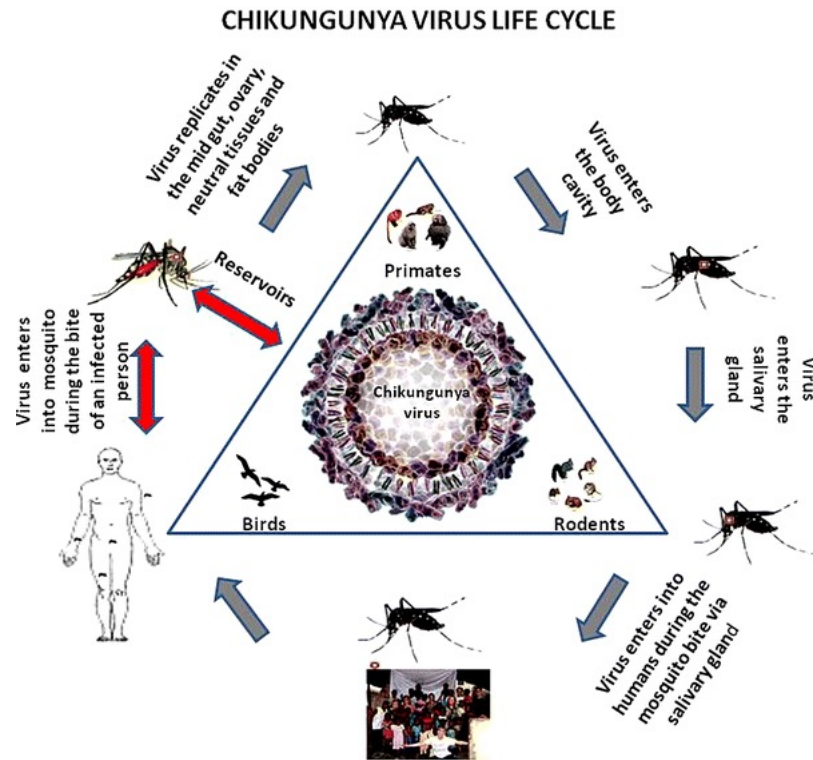
- Virus isolation (gold standard method)
- **Detection of –**
 - Virus-specific antibody
 - ELISA
 - Virus neutralization
 - Hemagglutination and hemagglutination inhibition tests
 - Viral nucleic acid
 - RT-PCR assay
 - Viral antigen in tissue samples
 - Fluorescent antibody staining
 - Immunohistochemical staining



Treatment and prevention

- There is no specific treatment for eastern equine encephalitis; clinical management is supportive
 - Horse with severe meningeal symptoms often require pain control and antiemetic therapy and rehydration for associated nausea and vomiting – close monitoring for the development of encephalitis
- In horse, inactivated vaccines against EEE and WEE viruses are commercially available
 - No eastern equine encephalitis virus vaccines are available for use in humans
 - Prevention of equine encephalitis depends on personal protective measures to decrease exposure to infected mosquitoes

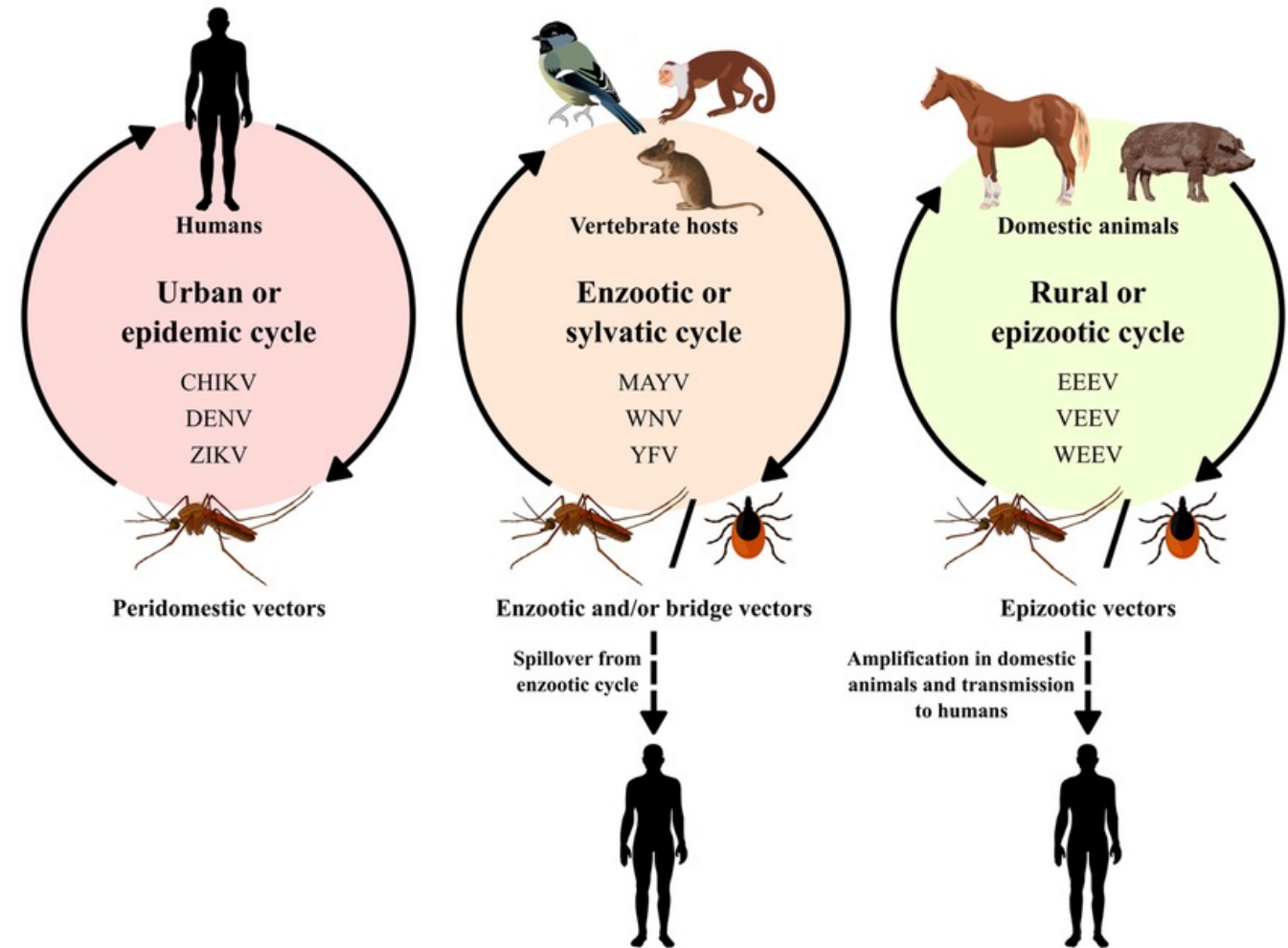
Chikungunya virus



- Animal hosts
 - Non-human primates, rodents, and birds
- Geographical distribution
 - Sub-Saharan Africa, India and Southeast Asia
 - Southern Europe
- Transmission cycle (urban CHIKV);
 - *Aedes aegypti*
 - *Aedes albopictis*
- CHIKV infection in humans usually causes a nonlethal, self-limiting, febrile illness
 - For acute infection; high fever, arthralgia, and rash are found, patients can experience painful and disabling arthritis that can last for 7 days, after the acute phase, polyarthritis can be recurrent and may persist for up to several years after infection
 - Other symptoms of CHIKV infection can include retro-orbital pain, neurological and hemorrhagic manifestations, and myocarditis

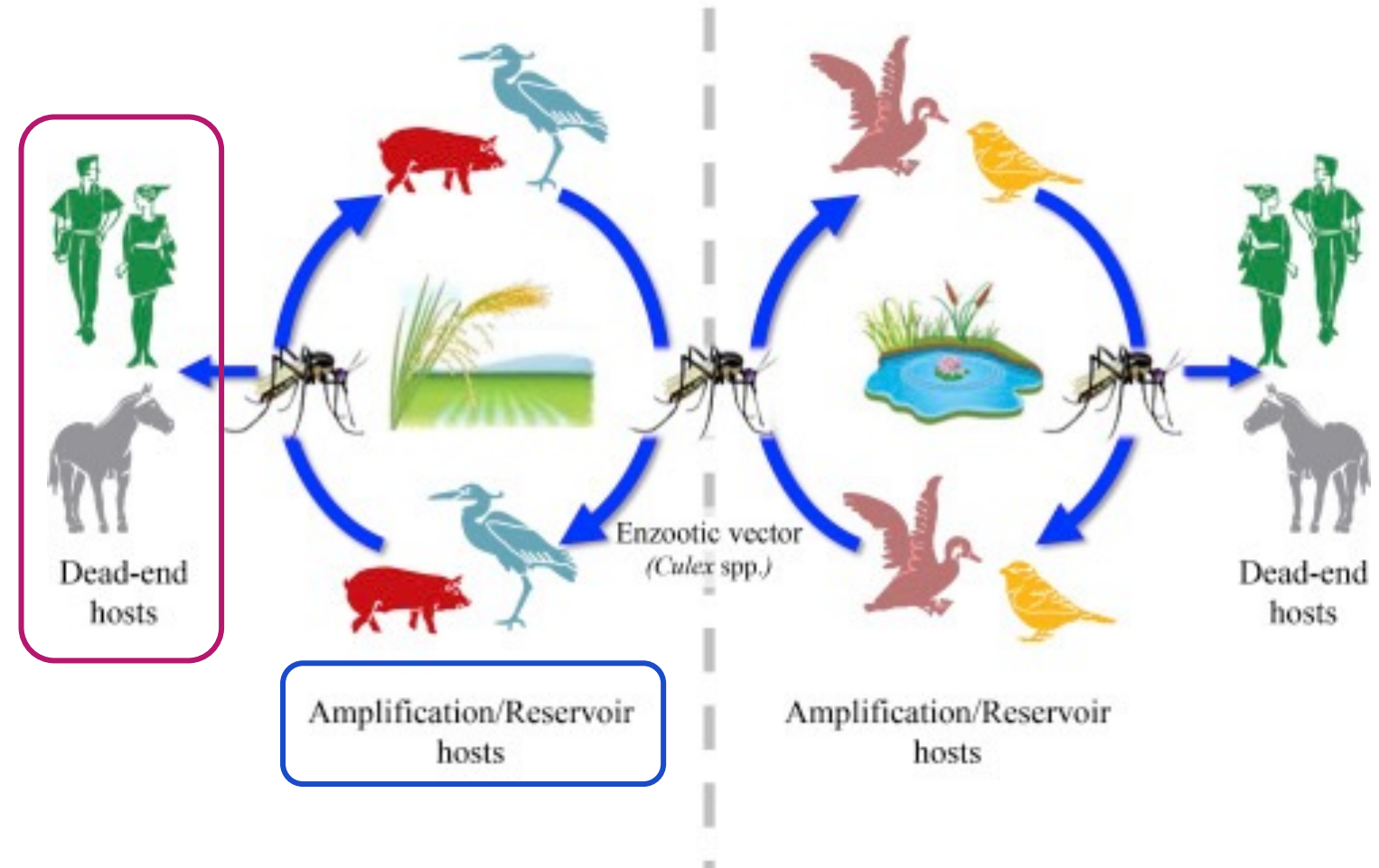
Family *Flaviviridae*

- Taxonomic structure of the family –
- Genus *Flavivirus***
 - Mosquito-Borne Flaviviruses
 - Japanese Encephalitis Virus
 - West Nile Virus
 - Yellow Fever Virus
 - Dengue Virus
 - Tick-Borne Encephalitis Virus
- Genus *Pestivirus***
 - Bovine Viral Diarrhea Virus
 - Classical Swine Fever Virus
- Genus *Hepacivirus*
 - Hepatitis C virus
 - Hepatitis G virus

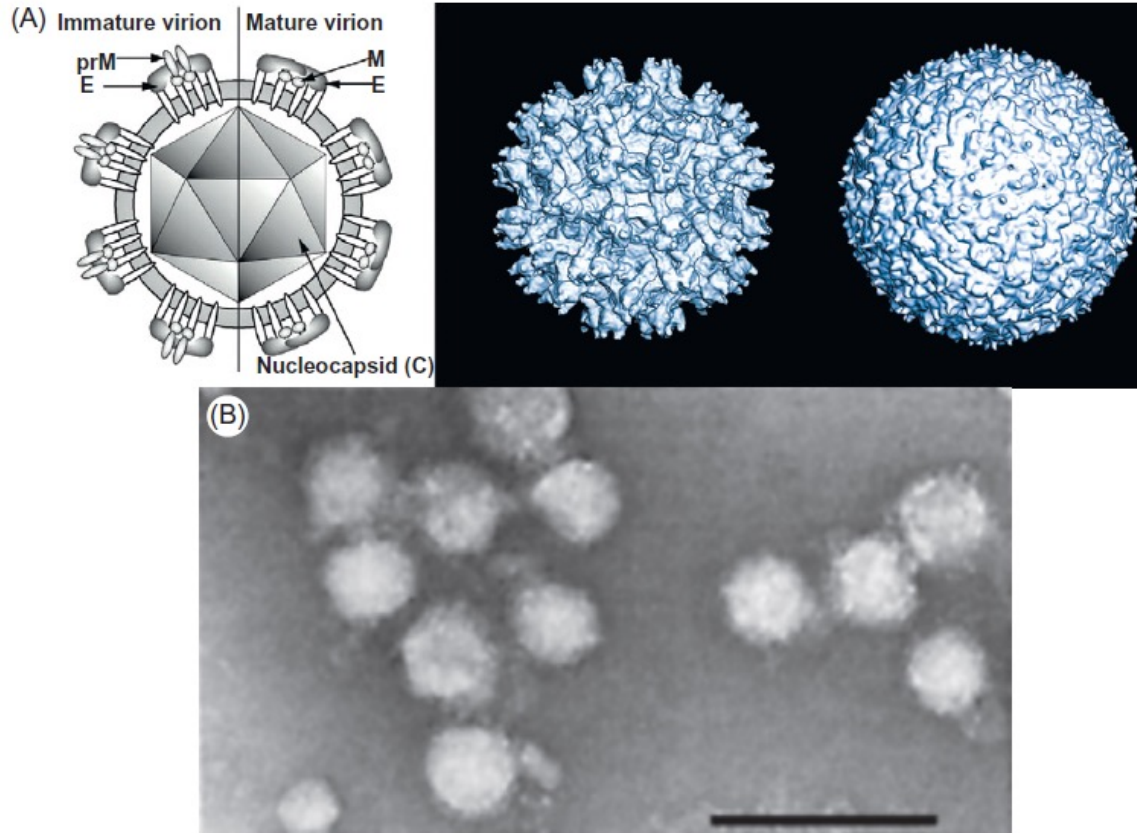


Japanese Encephalitis Virus

West Nile Virus



Flaviviruses
transmission
cycle

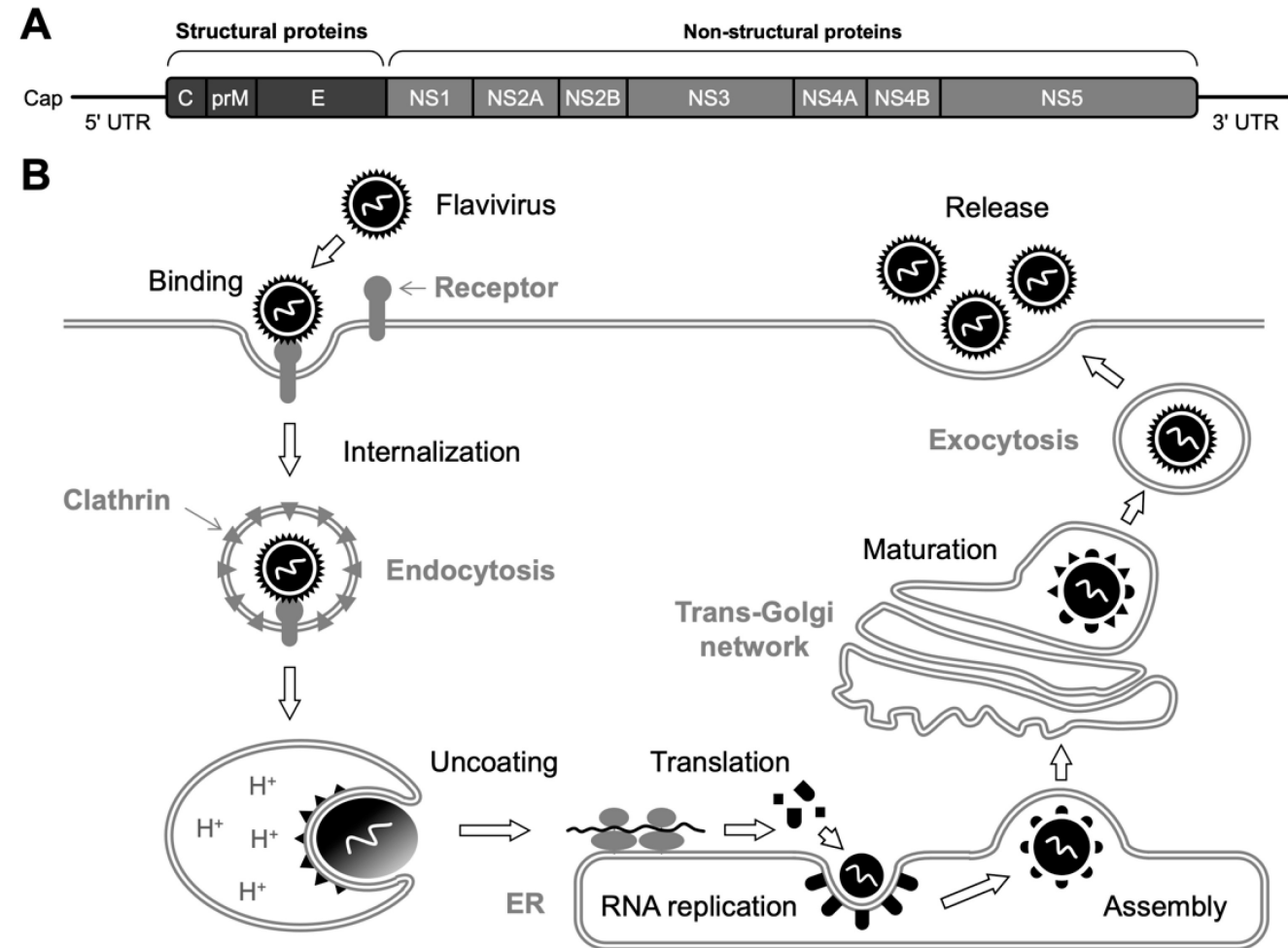


Virion properties

- Spherical shape with lipid bilayer envelope
- Size
 - Flaviviruses: 40–60 nm in diameter
 - Pestiviruses: 50 nm in diameter
- Icosahedron nucleocapsid (C)
- Envelope: glycoprotein spikes (mature virion – M and E proteins)
- Sensitive to heat and common disinfectants

Virus replication

- Genome
 - Single molecule of linear, positive-sense, single stranded RNA
 - Size:
 - Flaviviruses 11 kb
 - Pestiviruses 12.3 kb
 - Hepaciviruses 9.6 kb
- Cytoplasmic replication
- Maturation: intracytoplasmic membranes without budding



Disease in domestic animals and zoonotic disease in humans

VIRUS	RESERVOIR HOST	ARTHROPOD HOST	DISEASE	DISTRIBUTION
Genus Flavivirus				
Japanese encephalitis virus	Swine, humans, horses (birds)	Mosquitoes: Culex tritaeniorhynchus	Abortion, neonatal disease (encephalitis)	Asia
Murray Valley encephalitis virus	Humans (birds)	Mosquitoes: Culex annulirostris	Encephalitis	Aus, New Guinea
St. Louis encephalitis virus	Humans (birds)	Mosquitoes: Culex tarsalis, C. pipiens	Encephalitis	US, Canada, Central & South America
Wesselsbron virus	Sheep	Mosquitoes	Generalized infection, abortion	Africa

Disease in domestic animals and zoonotic disease in humans (cont.)

VIRUS	RESERVOIR HOST	ARTHROPOD HOST	DISEASE	DISTRIBUTION
Genus Flavivirus				
Dengue 1, 2, 3, 4 viruses	Humans (humans and monkeys)	Mosquitoes: Aedes aegypti, other Aedes spp.	Fever and rash, arthralgia, myalgia, hemorrhagic fever	Tropics worldwide
West Nile virus	Humans, horses, birds (birds)	Mosquitoes: Culex spp. (rarely ticks)	Fever, generalized Disease and Encephalomyelitis	Africa, Middle East, North Am., Central Am., South Am.
Tick-borne encephalitis virus	Humans (rodents, birds, ruminants)	Ticks: Ixodes spp. and via ingestion of raw milk	Encephalitis	Europe, Russia, Asia

Disease in domestic animals and zoonotic disease in humans (cont.)

VIRUS	RESERVOIR HOST	ARTHROPOD HOST	DISEASE	DISTRIBUTION
Genus Flavivirus				
Omsk hemorrhagic fever virus	Humans (muskrats)	Ticks: Dermacentor spp.	Hemorrhagic fever, gastrointestinal disease	Central Siberia
Kyasanur Forest disease virus	Humans (monkeys, rodents)	Ticks: Haemaphysalis spp.	Hemorrhagic fever, encephalitis	India (Mysore)
Powassan virus	Small mammals	Ticks: Ixodes spp.	Encephalitis	Canada, US, Russia
Louping ill virus	Sheep, horses, humans	Ticks: Ixodes ricinus	Encephalitis	Europe

Japanese Encephalitis Virus

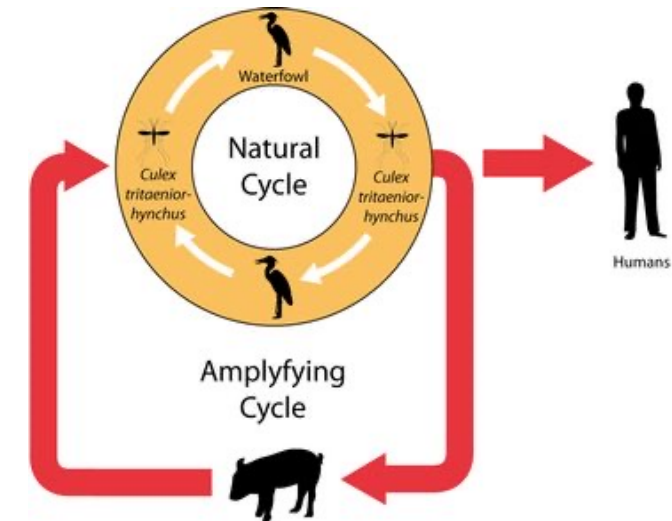


FIGURE 30.4 Distribution of Japanese encephalitis. (Courtesy of Centers for Disease Control.)

- Geographical distribution: Southeast Asia, Asia, Pacific islands, Northern Australia
- Primary mosquito vector: *Culex* mosquitoes especially *Culex tritaeniorhynchus*
- Important amplifying hosts: swine and birds
- Mosquito borne human pathogen –
 - 10 – 40% fatality rate
 - 40 – 70% permanent neurologic deficits

Animal hosts and clinical features

- Swine:
 - Natural hosts
 - Generally, inapparent
 - Stillbirths and abortions
- Horses: dead-end host
 - Fatal encephalitis
- Dogs, bats, reptiles found antibody against JEV
- Laboratory diagnosis: virus isolation, RT-PCR, antibody detection





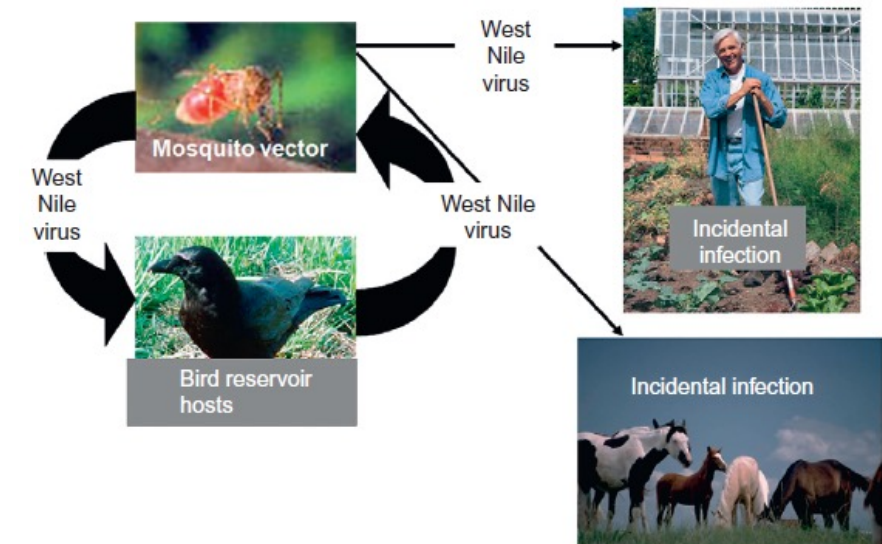
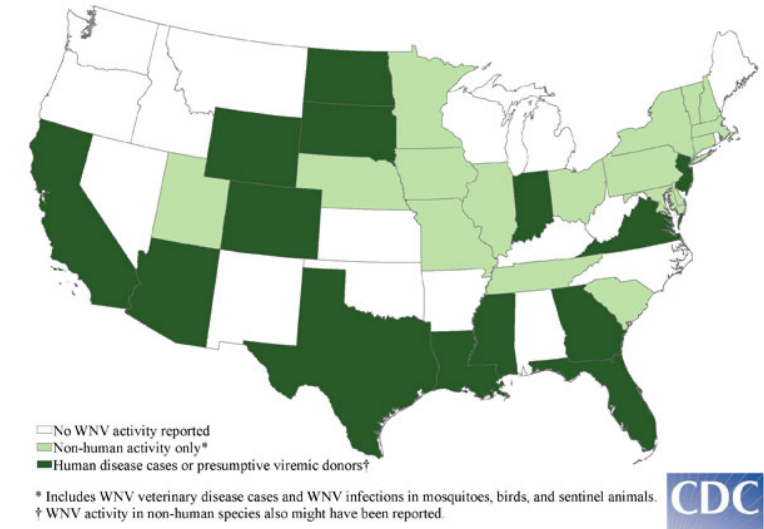
Treatment and control

- No specific treatment, clinical management is supportive
- Control: vaccination
 - Animals: swine, horses
 - Human especially children (JEV genotype III)



West Nile Virus

- Human infection
 - Mild febrile disease
 - Encephalomyelitis
- Distribution: Africa, Middle East, North America, Central America, and South America
- Transmission: Mosquito-wild bird-mosquito cycle





Clinical features in animals

- Cats, dogs, sheep, camelids, squirrels, and alligators
- Horses
 - Neurological signs
 - Depression
 - Abnormalities in gait
 - Head tremors
 - Weakness
 - Ataxia
 - Recumbency

Diagnostic techniques for JEV and WNV infection

- Serological tests
 - Virus-specific immunoglobulin M (IgM) antibodies
 - ELISA
 - Virus neutralization
- RT-PCR
- Immunohistochemistry

Bovine Viral Diarrhea Virus

- Genus *Pestivirus*: Bovine viral diarrhea viruses type 1 and 2
- Cytopathic viruses
 - Induce cytopathic effects in cultured cells
- Non-cytopathic viruses
 - Induce persistent infection in cells without cytopathic effects



Infection in animal

- Hosts: cattle, African and Indian buffalo, African antelope, New World camelids
 - Respiratory disease
 - Abortion and reproductive failure
 - Diarrhea
- Vertical transmission
 - Bovine embryo or fetus
 - Embryonic or fetal death
 - Teratogenesis
 - Persistent infection
 - Inapparent infection



Clinical Features

- Three situations:
 - Post-natal infection in non-pregnant cattle
 - Infection in pregnant cows
 - Persistent infection in calves and mucosal disease



Post-natal infection in non-pregnant cattle



- Most common in young animals
- Clinical signs: animal loss of antibodies
 - Incubation period: 5-7 days
 - Fever and leukopenia
 - Diarrhea
 - erosions or ulcers; lips, muzzle and within oral cavity
 - Thrombocytopenia
 - Hemorrhage
 - Immunosuppression
 - Increased incidence of opportunistic infections

Infection in pregnant cows

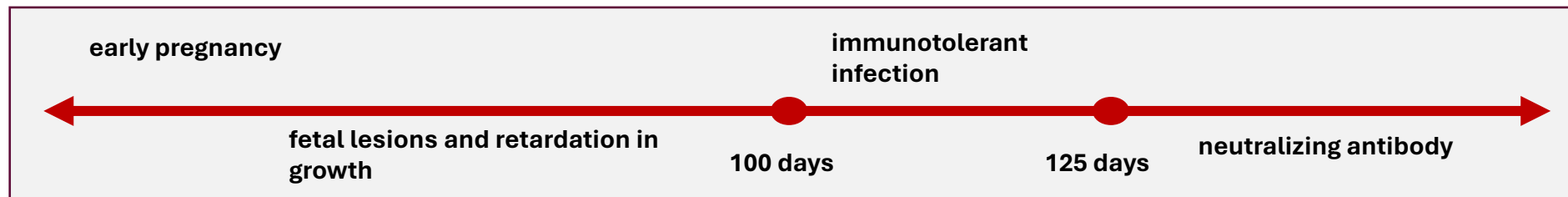
- Transplacental spread of virus to fetus
- Clinical signs depend on - age of fetus and strain of virus
 - Infection early in pregnancy
 - Embryonic death
 - Resorption
 - Infection before 100 days
 - Persistent postnatal infection
 - Fetal lesions and retardation in growth
 - Virus effects on organogenesis
 - Congenital defects
 - Retinal dysplasia
 - Cerebellar hypoplasia
 - Alopecia



Infection in pregnant cows (cont.)



- Infection at 100-125 days
 - Infected virus carriers for life
 - Never mount immune response to virus
 - Persistently immunotolerant infection
 - Seronegative in all standard tests
 - Shed virus in secretions and excretions
- Infection after 125 days
 - Neutralizing antibody



Persistent infection in calves and mucosal disease

- Mucosal disease
 - Non-cytopathic virus: persistently infection
 - Cytopathic virus
- Clinical signs:
 - Fever and anorexia
 - Watery diarrhea
 - Severe erosive or ulcerative stomatitis
 - Dehydration and emaciation
 - Death



FIGURE 30.8 Mucosal disease in a persistently infected bovine. Extensive and severe oral ulcers. (Courtesy of M. Anderson, University of California, Davis.)

Diagnosis

- Virus isolation in cell culture
- Detection of viral antigen in tissues
 - Immunofluorescent test
 - Immunohistochemical staining
- Detection of viral RNA in tissues or blood
 - RT-PCR
- Serological test

Classical Swine Fever Virus

- Classical swine fever or hog cholera
- Contagious disease of swine worldwide
- Transmission
 - Direct contact
- Control and prevention: vaccination, good hygiene



Clinical Features



รอยจำเลือดสีม่วงที่ผิวหนัง



ลูกสุกรนอนสุมกัน



ไทม์จุดเลือดออก

• Acute infection

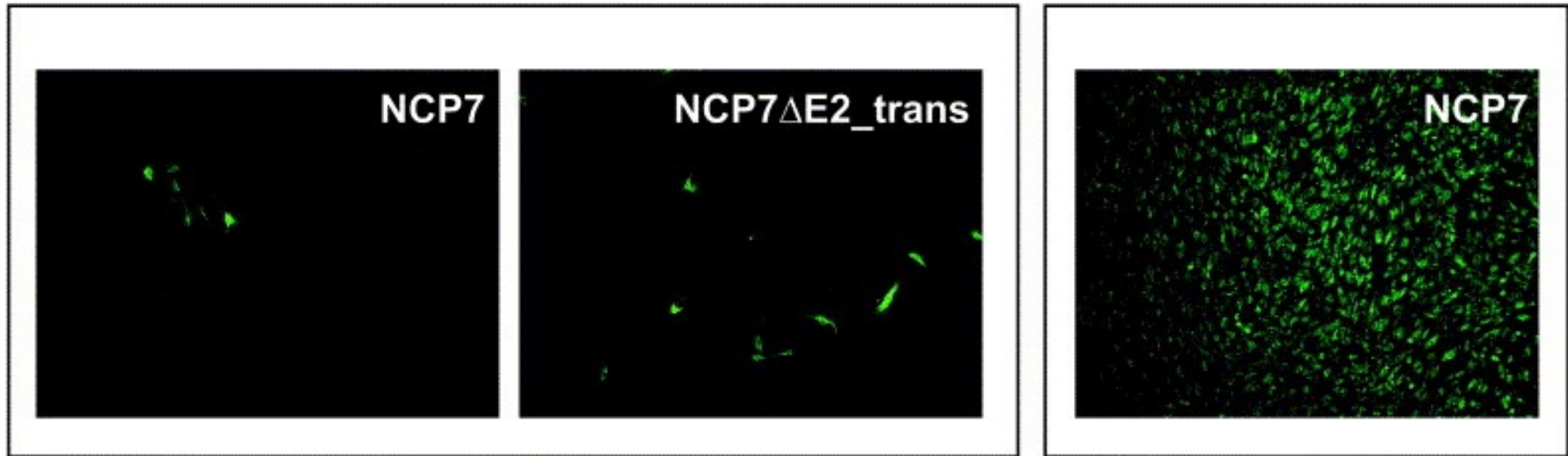
- Incubation period: 2-4 days
- High fever, depression, anorexia
- Conjunctivitis
- Vomiting and diarrhea
- Signs of nervous system dysfunction
 - Paresis, paralysis, circling, tremors, and convulsions
- Light-skinned swine
 - Diffuse hyperemia and purpura on abdomen and ears

• Chronic infection

- Prolonged incubation period
- Chronic diarrhea and dermatitis
- 2nd bacterial infections

Diagnosis

- Immunofluorescence or immunoperoxidase staining
- ELISA
- RT-PCR assays



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